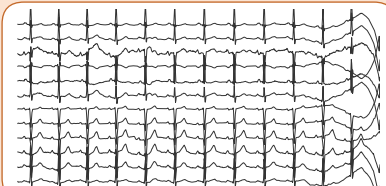


## A. Clinical ECG sources

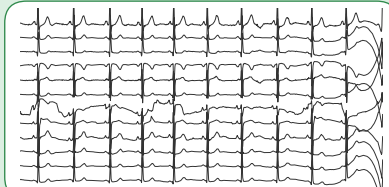
**PTB-XL**



**Georgia**



**Chapman**



**Ningbo**



### Harmonised label space

13 diagnostic labels retained  
across all sources | 66,379 recordings

### Leave-one-source-out evaluation

**Fold 1**

Test:

**PTB-XL**

**Fold 2**

Test:

**Georgia**

**Fold 3**

Test:

**Chapman**

**Fold 4**

Test:

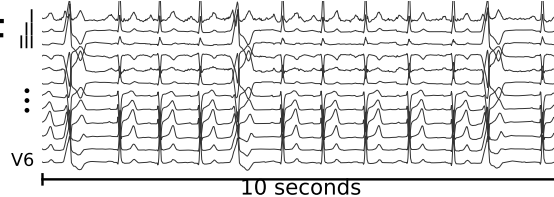
**Ningbo**

train on three sources, test on the fourth  
(four rotations)

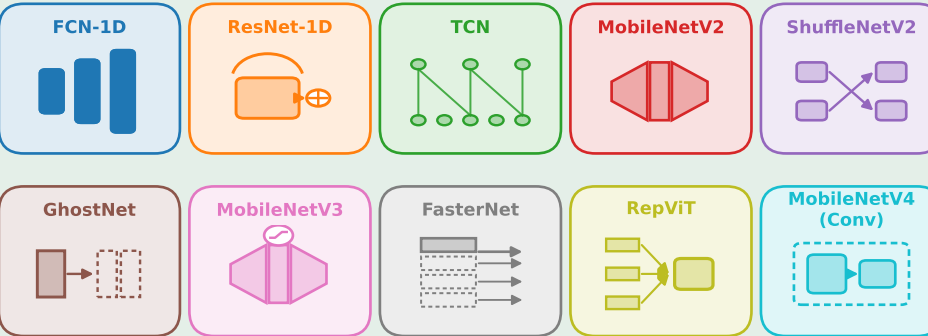
## B. Lightweight architecture benchmark

### 12-lead ECG input

10 s at 100 Hz  
(12 x 1,000 samples)



### 10 encoder families × 2 size budgets



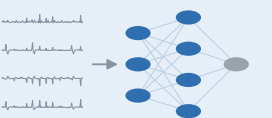
#### Compact

≤ 32 KB INT8 weights

#### Standard

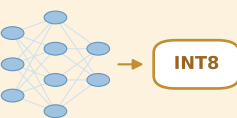
≤ 128 KB INT8 weights

### Train (FP32)



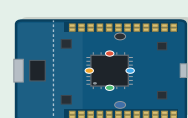
same protocol for  
all models and sources

### Quantise (INT8)



post-training  
quantisation

### Deploy (STM32F411)

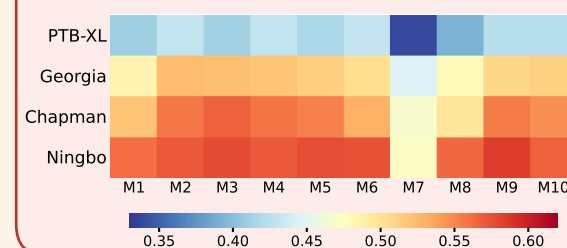


measured on device:  
Flash, SRAM, latency

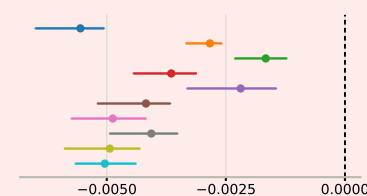
## C. Evaluation outputs

### RQ1. External discrimination

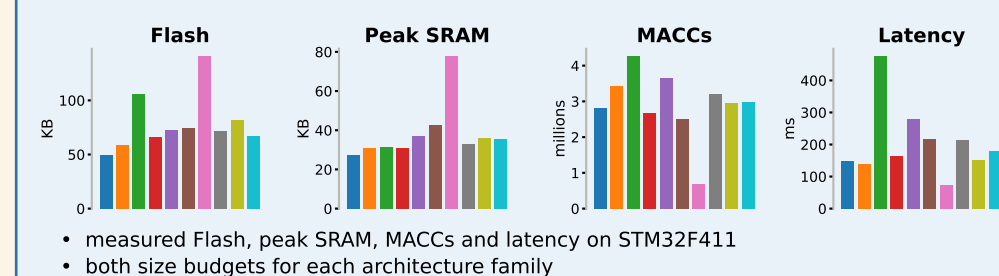
macro-AUPRC (FP32)



INT8 - FP32 (equal-source)



### RQ2. Hardware characterisation



### RQ3. Performance-resource trade-offs

